# Quantum Toom's Contours

David Ponarovsky

March 8, 2026

# Introduction

Today:

- Proof.
  1. Walktrough the math

# Our Code

## Code

Consider the representation of the $d$-dimensional toric by the cayley graph of the $(\mathbb{Z}_n^d, +)$ group generated by the elements $S = \{1_i : i \in [d]\}$ [1].

In our code, bits are set on the plaquette, and the edges correspond to checks that restrict the parity of the bits on their support to be even.

[1] where $1_i$ is the vector consisting of one at the $i$th coordinate and zero elsewhere

# Local Decoder

We say that an edge $\{x, y\}$ is positive relative to a plaquette $\{v, g_1 v, g_2 g_1 v, g_2 v\}$, defined by a rooted vertex $v \in \mathbb{Z}_n^d$ and the generators $g_1 \neq g_2 \in S$, if $x = v$ and $y$ is either $g_1 v$ or $g_2 v$.
We generalize Toom's rule in the following manner: Each plaquette asks if both of its positive edges point to a syndrome, namely unsatisfied. If so, then the bit flips itself; for convenience, we will think of the vertex at the end of the positive edges as the vertex which makes the decision and flips the plaquette.